

Continuous location of a service



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Managerial motivation

Introduction: Managerial motivation

- ▶ In this chapter we decide **where to place** a new facility on the plane
- ▶ The model is a **convex NLP**: the coordinates are continuous variables
- ▶ We will see why the choice of the objective is the real managerial decision

Where to place a service?

- ▶ A fast-charging station, a logistics micro-hub, a health-care post: we want to be “close” to demand
- ▶ But what does close mean?
 - ▶ *mean* distance: **efficiency**
 - ▶ *maximum* distance: **equity** towards the most unlucky user
 - ▶ *squared* distance: leads to the centroid
- ▶ Three objectives, three different points on the map: geometry makes the trade-off visible as in no other model

Efficiency versus equity

- ▶ Location decisions are among the most expensive ones to correct: a hub in the wrong place stays in the wrong place for years
- ▶ The continuous version (free coordinates on the plane) is the ideal laboratory for a conflict that reappears in all public and private decisions:
 - ▶ *efficiency*: serving well on average
 - ▶ *equity*: leaving nobody behind

In this chapter we will learn how to

1. formulate three location objectives and understand when they coincide
2. reformulate Euclidean distances with conic constraints and solve them with Gurobi from good starting points
3. handle convex geographic constraints
4. build an efficiency-equity frontier with the ε -constraint method



Summary: Managerial motivation

- ▶ Locating a facility means choosing its coordinates (x, y)
- ▶ “Close” may mean mean, maximum or squared distance
- ▶ The central theme is the conflict between **efficiency** and **equity**

Guided construction of the model

Introduction: Guided construction of the model

- ▶ We describe the problem in words, then the data and the variables
- ▶ We write two models: **Weber** (mean distance) and **minimax** (maximum distance)
- ▶ We add the quadratic variant and the geographic constraints
- ▶ Finally we see what characterizes the optimum of each objective

The problem in words

- ▶ *We decide* the coordinates at which the new facility is placed
- ▶ *The objective* — and this is the real managerial choice — may be to minimize:
 - ▶ the mean distance weighted by the users
 - ▶ the distance of the farthest user
 - ▶ the squared distance
- ▶ *The constraints*, when present, delimit the feasible area (a rectangular zone, a maximum radius from an infrastructure)

Data (input of the model)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of demand points (districts), indexed by $i \in \{1, 2, \dots, n\}$
(a_i, b_i)	$\in \mathbb{Q}^2$	coordinates of district i (km)
w_i	$\in \mathbb{Q}_{\geq 0}$	weight of district i : population, demand or priority
$(a_0, b_0), r$	$\in \mathbb{Q}^2, \in \mathbb{Q}_{>0}$	centre and radius of the possible geographic constraint (maximum distance from an infrastructure)

Decision variables

We introduce 2 free variables (they may take any real value) and, for the minimax version, a non-negative auxiliary variable:

$$\begin{cases} x = \text{east coordinate of the new facility (km)} \\ y = \text{north coordinate of the new facility (km)} \\ z = \text{distance from the farthest district (minimax only)} \end{cases}$$

Weber model (weighted mean distance)

$$\min \sum_{i=1}^n w_i \sqrt{(x - a_i)^2 + (y - b_i)^2} \quad (9.1a)$$

$$\text{subject to } x \geq 0, \quad (9.1b)$$

$$y \geq 0. \quad (9.1c)$$

- ▶ The convex objective function minimizes the total distance weighted by the districts (**efficiency**)
- ▶ The constraints define the variables, which are free

Minimax model

Minimax model (maximum distance)

$$\min z \quad (9.2a)$$

$$\text{subject to } \sqrt{(x - a_i)^2 + (y - b_i)^2} \leq z, \quad \forall i \in \{1, 2, \dots, n\}, \quad (9.2b)$$

$$x \geq 0, \quad (9.2c)$$

$$y \geq 0, \quad (9.2d)$$

$$z \geq 0. \quad (9.2e)$$

- ▶ The objective minimizes the distance of the farthest district (**equity**)
- ▶ The convex distance constraints impose that every district is at most z away from the facility (n constraints)
- ▶ The last constraints define the variables

Quadratic variant and geographic constraints

A third variant, quadratic: minimize

$$\sum_{i=1}^n w_i [(x - a_i)^2 + (y - b_i)^2],$$

which has a closed-form solution (the weighted centroid).

Typical **convex geographic constraints**, to be added to any variant:

- ▶ rectangular zone: $x^L \leq x \leq x^U, y^L \leq y \leq y^U$
- ▶ maximum distance from an infrastructure:
 $(x - a_0)^2 + (y - b_0)^2 \leq r^2$

Characterization of the optima (1/2)

Quadratic $\Rightarrow$ **centroid**. Setting the gradient to zero:

$$x^* = \frac{\sum_{i=1}^n w_i a_i}{\sum_{i=1}^n w_i} \quad (\text{and analogously for } y)$$

- ▶ It is the only case with a closed formula
- ▶ Useful as a starting point for the other models

Characterization of the optima (2/2)

Weber. It has no closed form; the function is not differentiable at the points (a_i, b_i) .

- ▶ At the optimum the weighted unit “pulls” of the districts cancel out: $\sum_{i=1}^n w_i \mathbf{u}_i = 0$
- ▶ $\mathbf{u}_i$ is the unit vector from the facility towards district i

Minimax. The optimum is the centre of the *smallest enclosing circle* of all the points:

- ▶ it depends only on the extreme districts and ignores the weights
- ▶ this is why it protects the outskirts

Summary: Guided construction of the model

- ▶ **Weber**: minimizes the weighted sum of Euclidean distances (efficiency)
- ▶ **Minimax**: minimizes the maximum distance with a variable z (equity)
- ▶ **Quadratic**: closed-form solution, the weighted centroid
- ▶ Geographic constraints (rectangle, circle) are convex and can be added to any variant

Worked example by hand

Introduction: Worked example by hand

- ▶ A one-dimensional example, solvable by hand
- ▶ We compute the optimum of the three objectives on the same data
- ▶ We will see that the three objectives give three different answers

Example: Three customers on a road (1/2)

Customers at km 0, 4 and 10 with weights 1, 1 and 3.

Weber in 1D = weighted median. The cost is

$$f(x) = |x| + |x - 4| + 3|x - 10|.$$

- ▶ Moving by δ to the left from $x = 10$, the cost of the first two customers decreases by 2δ but that of the third increases by 3δ : it gets worse
- ▶ From the right, symmetrically, it gets worse again
- ▶ Optimum: $x^* = 10$, the point that leaves *at least half of the total weight* ($5/2 = 2.5$) on both sides: the weighted median

Example: Three customers on a road (2/2)

Centroid.

$$x = \frac{1 \cdot 0 + 1 \cdot 4 + 3 \cdot 10}{5} = 6.8$$

sensitive to *how far* the points are, not only to which side they are on.

Minimax. Midpoint of the extremes: $(0 + 10)/2 = 5$, weights ignored.

Three reasonable objectives, three different answers: 10, 6.8, 5.
The choice of the objective **is** the managerial decision.

Summary: Worked example by hand

- ▶ In 1D Weber coincides with the **weighted median**
- ▶ The centroid is the **weighted mean**: it depends on how far the points are
- ▶ The minimax is the **midpoint of the extremes**: it ignores the weights

Case study

Introduction: Case study

- ▶ We apply the models to a city with twelve districts
- ▶ We see the modelling trick that makes Euclidean distances tractable for Gurobi: conic constraints

The data of the case

- ▶ Twelve districts with weights between 5 and 25 thousand inhabitants (file `localizzazione_quartieri.csv` in the folder `data/`)
- ▶ Technical constraint: the station must lie within $r = 2$ km of the substation at $(7, 6)$

The modelling trick: conic constraints

- ▶ A variable $d_k \geq$ the Euclidean distance from district k , imposed with the (convex) conic constraint

$$d_x^2 + d_y^2 \leq d_k^2$$

- ▶ Weber minimizes $\sum_k w_k d_k$
- ▶ The minimax minimizes a variable z with $d_k \leq z$

The model in Gurobi

```
d = m.addVars(n) #  $d_k \geq \text{dist. from district } k$ 
for k in range(n):
    m.addConstr(dx[k] == px - coord[k, 0])
    m.addConstr(dy[k] == py - coord[k, 1])
    m.addQConstr(dx[k]**2 + dy[k]**2 <= d[k]*d[k]) # cone
# Weber:  $\min \sum w_k d_k$ 
# minimax:  $\min z$  with  $d_k \leq z$ 
# geographic constraint:  $(px-7)^2 + (py-6)^2 \leq R^2$ 
```

Summary: Case study

- ▶ The Euclidean distance is modelled with a variable d_k and a convex **conic constraint**
- ▶ Weber and minimax differ only in the objective function
- ▶ The substation constraint is a circle: a convex quadratic constraint

Results

Introduction: Results

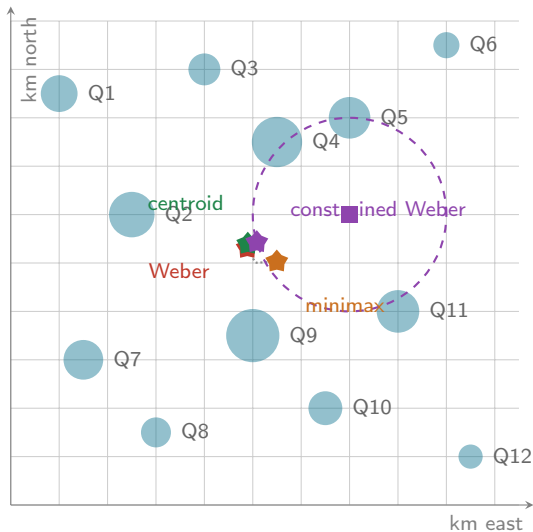
- ▶ We compare the four optimal locations: centroid, Weber, minimax, constrained Weber
- ▶ We see them on the map of the city
- ▶ We draw a managerial lesson on the cost of constraints

The four solutions

Solution	(x, y)	mean dist.	max dist.
Weighted centroid	(4.897, 5.397)	–	–
Weber	(4.886, 5.310)	3.344 km	6.314 km
Minimax	(5.496, 5.029)	3.391 km	5.680 km
Constrained Weber	(5.083, 5.429)	cost +0.2%	

- Constrained Weber: the constraint is active, the optimum lies on the boundary of the circle

Map of the city



How to read the map

- ▶ The teal circles are the 12 districts (area proportional to the population)
- ▶ The stars indicate the four optimal locations: Weber in red, minimax in orange, centroid in green, constrained Weber in purple
- ▶ The dashed purple circle is the constraint of 2 km from the electrical substation
- ▶ The dotted line is the trajectory of the compromise solutions between minimax and Weber

Managerial insight

- ▶ Weber and the centroid almost coincide here (the weights are evenly distributed)
- ▶ The minimax moves more than half a kilometre towards the south-east in order to protect the peripheral districts
- ▶ The constraint of the substation costs very little ($+0.2\%$ of total distance)
- ▶ Discovering that a feared constraint is almost free is a managerial result as important as the optimum itself

Summary: Results

- ▶ Weber $\approx$ centroid when the weights are balanced
- ▶ The minimax moves towards the outskirts
- ▶ The geographic constraint is active but costs only 0.2%

Sensitivity analysis: the efficiency-equity frontier



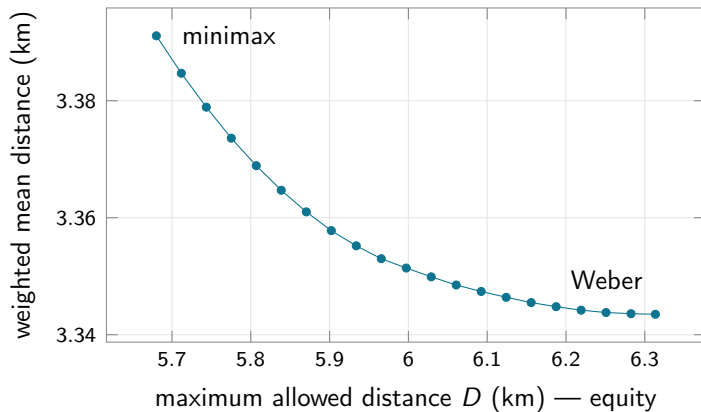
Introduction: the efficiency-equity frontier

- ▶ Weber and minimax are two extremes: are there intermediate solutions?
- ▶ We build them with the ε -constraint method
- ▶ The shape of the frontier tells how much equity costs

The ε -constraint method

- ▶ We minimize the mean distance while imposing a cap D on the maximum distance
- ▶ We vary D between the two extremes:
 - ▶ $D = 5.680$ km: minimax solution
 - ▶ $D = 6.314$ km: Weber solution
- ▶ For every D we obtain the minimum attainable weighted mean distance

Efficiency-equity frontier



Some points of the frontier

Cap D	Mean distance	
5.680 km	3.391 km	minimax solution
5.997 km	3.351 km	
6.314 km	3.344 km	Weber solution

- ▶ The extremes of the curve are the minimax solution (on the left) and the Weber point (on the right)
- ▶ The almost flat curve indicates that equity costs very little in terms of efficiency

An almost flat frontier

- ▶ Guaranteeing 0.63 km less to the farthest district costs only 47 metres of mean distance
- ▶ When the frontier is this flat, the equitable solution is “almost free”: a very strong argument in a public discussion
- ▶ The opposite case (a steep frontier) would signal a genuine efficiency-equity conflict

Summary: the efficiency-equity frontier

- ▶ The ε -constraint method optimizes one objective and caps the other
- ▶ The frontier connects minimax and Weber
- ▶ Flat frontier: equity is almost free; steep frontier: genuine conflict

Exercises

Introduction: Exercises

- ▶ Five exercises to test the models of the chapter
- ▶ Checks by hand, sensitivity to the weights, extensions of the model (two facilities, Manhattan distance, multipliers)

Exercise: check

- ▶ Compute by hand the weighted centroid of the 12 districts from the CSV data
- ▶ Compare it with the output of the script

Exercise: a weight that changes everything

- ▶ Raise the weight of Q12 (the district at the bottom right) from 5 to 40
- ▶ How far do Weber and minimax move?
- ▶ Which of the two is more sensitive to the weights, and why?

Exercise: two facilities

- ▶ With *two* stations and the assignment of every district to the nearest one, does the problem remain convex?
- ▶ Try an alternating approach (assign–optimize)
- ▶ Discuss why it only finds local optima

Exercise: Manhattan distance

- ▶ Replace the Euclidean distance with $|x - a_i| + |y - b_i|$
- ▶ Show that the Weber problem becomes an LP (introduce auxiliary variables for the absolute values)
- ▶ Solve it with Gurobi and compare the optimum

Exercise: multiplier of the radius

- ▶ Estimate by perturbation the multiplier of the constraint $r = 2$ km (solve with $r = 2.1$)
- ▶ How much is every additional 100 m of radius worth, in km of total distance?

Summary: Exercises

- ▶ The exercises check the centroid, the sensitivity to the weights and the value of the geographic constraint
- ▶ The extensions show when the problem loses convexity (two facilities) or becomes an LP (Manhattan)

Summary (1/2)

Models:

- ▶ Weber (weighted mean distance), minimax (maximum distance), quadratic (centroid)
- ▶ Euclidean distances are written with convex conic constraints

Optima:

- ▶ Centroid: closed formula; Weber: the weighted pulls cancel out; minimax: centre of the smallest enclosing circle

Summary (2/2)

Managerial lessons:

- ▶ The choice of the objective is the managerial decision
- ▶ A feared constraint can be almost free (+0.2%)
- ▶ The efficiency-equity frontier measures how much equity costs:
here almost nothing